TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Dr.-Ing. Jakob Köchermann	Email	jakob.koechermann@igb.fraunhofer.de
Institute Name	Fraunhofer Center for Chemical-Biotechnological Processes CBP			
Address	Am Haupttor Bau 1251, 06237 Leuna, Germany			

2. Technology Overview

Title of Proposed Technology	*Flexible Power-to-Methanol System Operation (SynFlex)*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (X), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		*Energy & Resources*	
	Medium Category	Small Category	Medium Category	Small Category
	Greenhouse Gas Processing	*CO2 Conversion Technologies*	New & Renewable Energy	Synthetic Fuels
Concept of Proposed Technology	*The proposed technology enables flexible, dynamically operated methanol synthesis using green hydrogen and captured CO_2 within an isothermal, high-pressure reactor system. Two reactors in series, each with intermediate condensation and full recycle, allow precise control of thermal management and conversion even under rapidly changing operating conditions. By experimentally exploring load changes, start-up/shut-down behavior, and transient system responses, the project demonstrates how methanol production can be integrated into fluctuating renewable-energy environments. The resulting insights support the design of robust Power-to-X systems and help industry partners assess the feasibility of flexible, sector-coupled methanol synthesis.*			
Rationale for Joint R&D Project	*The joint R&D project is essential because flexible methanol synthesis requires a catalyst that remains active and stable under rapidly changing operating conditions, which cannot be developed or validated by a single partner alone. Integrating the process into a renewable-based hydrogen infrastructure demands close collaboration between catalyst experts, reaction engineers, and system integrators to understand transient behavior and ensure compatibility with fluctuating power supply. Moreover, different CO_2 qualities and sources—ranging from technical process gases to biogenic streams—must be systematically evaluated to guarantee robust operation across*			

	real-world feedstock variations. Only a coordinated, interdisciplinary effort can generate the data, models, and technological readiness required for scalable, industry-relevant Power-to-Methanol solutions.
Key Words	1. *Dynamic Power-to-Methanol Operation* 2. *Hydrogen Integration* 3. *Catalyst Stability* 4. *CO_2 Flexibility* 5. *Dynamic Process Simulation*

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (6)	End of the Project	Level (6)
R&D Objectives	*The project aims to develop technologies that enable highly flexible, load-responsive methanol synthesis under fluctuating hydrogen and CO_2 supply. This includes advanced reactor control strategies capable of maintaining stable isothermal conditions during rapid changes in feed composition and throughput. A dedicated catalyst activation and conditioning technology will be established to ensure reliable performance under dynamic operation. In addition, dynamic simulation tools will be developed to predict transient reactor behavior, guide testing strategies, and support process optimization. Together, these technologies create a robust platform for renewable-integrated, industrially scalable Power-to-Methanol operation.*			
R&D Target Specifications	*The project aims to show that the methanol plant can operate stably between 20% and 100% load, including both fast step changes and slow, controlled ramps. To check this, we will measure load level, ramp rate (e.g., % per minute), and how pressure, temperature, and gas flow change inside the reactor during these transitions. A dynamic reactor model will be developed and validated using these measured values, including temperature peaks, pressure trends, and flow behavior. The project will also generate detailed datasets such as gas composition, conversion rates, and energy use, which can be used to design or validate future process models.*			
R&D Plan	***Phase I – Pre-Study (6 months)*** *Phase I focuses on establishing the technical basis for flexible methanol synthesis, including defining operating limits, load-variation scenarios, and measurement strategies. During this phase, the pilot plant setup, data-acquisition tools, and safety procedures are reviewed and validated to ensure reliable dynamic testing. A preliminary assessment of reactor behavior under selected load changes is performed, and requirements for the dynamic reactor model are specified. This phase concludes with a validated test plan and a clear modelling framework for Phase II.* ***Phase II – Main R&D (30 months)*** *Phase II centers on executing systematic dynamic load experiments across the 20–100% operating range, including both rapid step changes and controlled ramp profiles. Comprehensive data on pressure, temperature, gas composition, and flow behavior are collected to evaluate system performance and feed the development of a predictive dynamic reactor model. Iterative modelling and experimentation cycles refine understanding of reactor response, hotspot formation, and process controllability. The phase concludes with a validated dynamic model, a complete dataset for future process design or simulation, and an overall assessment of flexible methanol production under real-world conditions.*			

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*The Fraunhofer CBP will act as the central experimental hub, operating its pilot-scale methanol synthesis plant to carry out dynamic load tests and system characterization under real-world conditions. Its responsibilities include running and supervising the reactors, ensuring safe operation, and providing comprehensive process analytics through its in-house analytical capabilities. CBP will supply the necessary infrastructure, measurement tools, and technical staff, including resources for hydrogen handling and continuous plant monitoring. It will also generate detailed datasets on temperature, pressure, gas composition, and flow behavior to support modelling and performance evaluation. Finally, CBP will contribute its process expertise to interpret results and ensure that the technology can be scaled toward industrially relevant applications.*
Expected R&R of Korean SME Partner	*The Korean SME(s) may contribute in a **modular and optional manner**, taking responsibility for one or several technical tasks depending on their expertise and capacity. Possible roles include acting as a **catalyst provider**, supplying and evaluating methanol-synthesis catalysts under dynamic conditions. SMEs may also support **data generation** for the dynamic reactor model developed by Fraunhofer ITWM, for example through lab-scale testing or kinetic measurements. Additional contributions may involve **system-integration activities**, such as assessing how flexible methanol synthesis can be linked to renewable-energy operation and hydrogen supply concepts. Furthermore, SME partners may explore **methanol downstream applications** - including conversion to jet fuel, synfuels, marine fuels, or chemical intermediates - to connect the project with broader Power-to-X value chains.*

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*The project is expected to deliver **technological breakthroughs in flexible methanol synthesis**, including validated reactor operation across dynamic load ranges and high-quality datasets for modelling, control, and system design. It will establish a **predictive framework** for integrating methanol production with fluctuating renewable energy inputs and variable hydrogen supply, strengthening future Power-to-X value chains. The outcomes will also include **improved understanding of catalyst performance** under real-world dynamic conditions and insights into downstream methanol applications such as sustainable fuels or chemical intermediates, which are increasingly relevant for a decarbonized industry. In terms of commercialization, the project will reduce technological risk, provide scalable operational strategies, and support the development of economically viable renewable methanol pathways. Ultimately, the work will contribute to **industrial competitiveness**, emissions reduction, and economic opportunities in emerging e-fuel and sustainable chemical market*

6. References

(1) E-CO_2Met – Electricity & CO_2 to Methanol (https://www.cbp.fraunhofer.de/en/reference-projects/e-co2met.html)

(2) S. Rönsch, J. Köchermann, J. Schneider, S. Matthischke, Global Reaction Kinetics of CO and CO_2 Methanation for Dynamic Process Modeling, Chem. Eng. Technol. 39 (2016) 208–218.

(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

(max 10 pages)